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Thermal analysis of high frequency electromagnetic heating of lossy porous media

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HIGHLIGHTS

- Analytical solutions are presented for electromagnetic heating of lossy media.
- The solutions can be used as forward models in design of experimental studies.
- The solutions allow estimation of electromagnetic absorption coefficient.
- The solutions allow estimation of electromagnetic power loss.

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ABSTRACT

Industrial applications of the electromagnetic wave as a source of energy have been increasing in recent years. One of the growing areas of application of electromagnetic heating (EMH), especially using high frequency waves, is the heating of geological media for extraction of hydrocarbons. In this work, electromagnetic field propagation in a lossy, electrically isotropic, and homogenous medium was studied to determine the effectiveness of EMH for heavy oil recovery. The electromagnetic heat source is modeled using Beer-Lambert's law coupled with heat conduction. New analytical solutions are developed to find temperature distribution in linear and radial geometries subject to various boundary conditions. The effect of electromagnetic heat generation on thermal penetration depth is examined. One of the challenges in electromagnetic heating of geomaterials has been reflection of the incident wave due to impedance mismatch and thus loss of the applied power. The analytical solutions developed allow determination of absorbed power by the geological media and thus can be used as a tool to predict efficiency of electromagnetic heating. In addition, the aforementioned solutions can be used as forward model for design of experiments and to estimate the electromagnetic absorption coefficient of geomaterials.

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1. Introduction

Electromagnetic heating (EMH) is a generic term for thermally stimulating materials with electrical energy. It can be divided into four main distinct groups; (i) direct resistance heating (DRH); (ii) induction heating (IH) for heating highly conductive materials; (iii) radiofrequency heating (RFH); and (iv) microwave heating (MWH) for relatively low-conductivity materials. At low frequencies (<100 Hz) resistance heating or so called Joule or Ohmic heating controls the process. At high frequencies between radio frequency (between 10 and 100 MHz) and microwave frequency (between 100 MHz and 100 GHz) dielectric heating dominates (Basak and Ayappa, 1997). A household microwave oven operates at 2.45 GHz.

Dipole polarization is the mechanism by which high frequency electromagnetic waves heat poor conductors or dielectrics. When employing high frequency EM energy, polar molecules, such as water molecules having an electrical dipole moment, generate heat as a result of molecular rotations caused by oscillations. Therefore, the heat source of high frequency heating is volumetrically distributed and located in situ and thus is more easily conveyed as compared to the conduction or convection based heating methods, wherein the heating source is applied at the boundaries of the medium (Hassanzadeh and Harding, 2016).

The determination of electromagnetic heating efficiency of geomaterials needs solution of coupled energy and wave equations. Maxwell's equations are generally used to describe the wave

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Nomenclature

Roman letters

A	cross sectional area (m^2)
$\vec{a}$	unit vector (–)
$\vec{B}$	magnetic flux density (Wb/m^2)
$\vec{D}$	electric flux density (C/m^2)
$\vec{E}$	electric field intensity (V/m)
E_0	magnitude of electric field intensity at inlet (V/m)
H	magnetic field intensity (A/m)
H_0	magnitude of electric field intensity at inlet (A/m)
H_0^1	Hankel's Bessel function of the first kind (–)
h	thickness (radial) (m)
$\vec{J}$	electric current density (A/m^3)
j	complex index (–)
k_T	effective thermal conductivity (W/m K)
L	length (linear) (m)
M	magnetization of the material (A/m)
M_t	volumetric heat capacity (J/m^3 K)
N	Stark Number (–)
$\vec{P}$	dielectric polarization of the material (C/m^2)
P_0	power (W)
$\dot{q}_{em}$	volumetric heat generation (W/m^3)
r	r-direction (Radial) (m)
r_w	inner radius (m)
r_e	outer radius (m)
T	temperature (K)
t	time (s)
V	volume (m^3)
x	x-direction (Cartesian) (m)
y	y-direction (Cartesian) (m)
z	z-direction (m)

Greek letters

α_{em}	attenuation factor (or absorption coefficient) (1/m)
β_{em}	phase constant (or wavenumber) (1/m)
ε_0	electric permittivity of the vacuum (F/m)
ε	electric permittivity of the medium (F/m)
η	intrinsic impedance of the media (Ω)
ϕ	azimuthal angle direction (Radial) (–)
γ_{em}	propagation constant (1/m)
μ_0	magnetic permeability of the vacuum (H/m)
μ	magnetic permeability of the medium (H/m)
ω	angular frequency (rad/s)
$\vec{\wp}$	Poynting vector (W/m^2)
ρ_v	volume charge density (C/m^3)
σ	medium conductivity (S/m)
θ_η	impedance phase (degree)
τ	period (s)
χ	electric susceptibility of the medium (–)

Subscripts

D	dimensionless parameter (–)
em	electromagnetic (–)
s	phasor form (–)

propagation (Sadiku, 2014). They can be simplified to Beer-Lambert's law (Hippel, 1954) using several assumptions. Beer-Lambert's law states that the power and field intensity decreases exponentially as the wave propagates through the medium. The law is appropriate for semi-infinite media (Ayappa et al., 1991a,b).

EMH has applications in many industries including the polymer and wood manufacturing industries, the food industry, biotechnology and medical fields, and in oil extraction. The food industry is the main user of microwave heating for fast food preparation and rapid heating of the foods. While highly effective and ubiquitous, microwave heating still suffers from the existence of hot/cold spots or uneven heat distribution. There is no such deficiency for radio frequency heating (Hossan and Dutta, 2012). Several studies have been conducted for modeling of the EM heating process of food. Ayappa et al. (1991a,b) compared the Beer-Lambert power formulation with the numerical solution of Maxwell's equations. They found that there is a critical slab thickness above which the Beer-Lambert law limit is valid. Yang and Gunasekaran (2004) used pulsed and continuous microwave energy to heat cylindrical gelatinous substances. Their analysis showed that the Beer-Lambert law resulted in comparable accuracy to Maxwell's equations.

Short processing time and quick start-up of electromagnetic heating led to significant popularity in material synthesis processes. Bhattacharya et al. (2013) optimized the heating process by using four different electromagnetic source configurations. They showed that heating pattern is critical to homogenize the reactor dynamic and reduce the reaction time. Electromagnetic heating can also be applied in fabrication processes, especially in bonding polymer-based devices by heating a highly dielectric thin material sandwiched between two layers (Lei et al., 2004; Rahbar et al., 2010). In biotechnology, the pros and cons of applying electromagnetics have been investigated in recent years. The health risks due to near-field (wireless communication center) exposure was the main focus of some studies (Keangin et al., 2011). Also, extensive numerical and experimental studies have been done on the microwave ablation (MWA) process, which has been used for neutralizing cancer cells (Rattanadecho and Keangin, 2013). In the wood manufacturing industry, microwave irradiation has been used for thawing frozen wood, drying or killing microorganisms and insects (Erchiqui, 2013).

In heavy oil recovery, the use of EMH in conjunction with solvent injection to dilute the bitumen is increasing due to the high energy intensity of the current thermal recovery methods and the associated environmental footprint. Abernethy (1976) developed an elegant analytical model based on the Beer-Lambert power formulation to find the temperature distributions from the radiation of electromagnetic energy in a radial system. However, the full coupling of heat conduction and the Beer-Lambert power formulation was not possible. Steady-state flow performance of a vertical well was compared to an unheated case. In another modeling effort, an algorithm was developed by Fanchi (1990) for estimating the temperature increase associated with reservoir electromagnetic heating. An asymptotic solution to Maxwell's equations was presented and it was shown that for an axially symmetric wave, which is propagating radially, the power attenuation follows the Beer-Lambert law. Carrizales et al. (2008) developed analytical solutions for single-phase and steady-state counter-current flow where the source antenna is placed in the production well and for the case of co-current flow where the flow is taking place in the same direction as the electromagnetic wave propagation.

In this study, our focus is on the EMH of heavy oil reservoirs by installation of an electromagnetic antenna in the SAGD-style well-pairs. At the beginning of the SAGD-type processes, the fluid around the wellbores is immobile and thus preheating is required to establish the fluid communication between the two wells. In this stage, the dominant mechanisms of heating are conduction

and radiation. By using simplifying assumptions for wave propagation through the medium, we derived analytical solutions for temperature profile in a homogeneous reservoir. These assumptions have been extensively used in the literature and very common when an analytical solution is sought. The developed analytical solutions are used to obtain asymptotic solutions that allow prediction of the absorbed power and average absorption coefficient of the target zone. The late-time solutions show that the collected temperature data of preheating process in field projects or laboratory experiments can be analyzed to estimate electrical properties of the porous media.

Such solutions are required for more advanced understanding of the physical problems and validation of more sophisticated numerical models. The practical value of these solutions lies in the fact that they aid to improve interpretations and conduct fast sensitivity analysis and computations. To the best of our knowledge, these analytical solutions that are essential in application to EM heating of materials have not been previously reported in the literature.

2. Mathematical model

Although the energy transfer mechanisms of different types of EMH methods are completely different, it is possible to model them using a similar theoretical approach. The coupling of electromagnetic wave propagation and heat transport is necessary to study EMH in lossy geological media.

2.1. Electromagnetic waves propagation in lossy media

A lossy dielectric is a partially conducting medium (imperfect dielectric or imperfect conductor), which absorbs energy of the electromagnetic wave as it propagates inside the material. Electromagnetic wave propagation can be described by Maxwell's equations (Sadiku, 2014)

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \tag{1}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \tag{2}$$

$$\nabla \cdot \vec{D} = \vec{\rho}_v \tag{3}$$

$$\nabla \cdot \vec{B} = 0 \tag{4}$$

where $\vec{E}$ and $\vec{H}$ are electric and magnetic field intensities in V/m and A/m, respectively. The parameters $\vec{D}$ and $\vec{B}$ are electric and magnetic flux densities in C/m^2 and Wb/m^2. The two parameters $\vec{\rho}_v$ and $\vec{J}$ are the volume charge density and electric current density as external sources in the medium or the conductivity of the medium in the absence of the induced polarization charge or current measured in C/m^3 and A/m^3. Eq. (1) is Faraday's law of induction. Eq. (2) is Ampere's law modified by Maxwell to include displacement current ($\partial \vec{D}/\partial t$) for prediction of electromagnetic wave propagation, and the third and fourth equations are Gauss' laws for the electric and magnetic fields, respectively. The volume charge and electric current densities, $\vec{\rho}_v$ and $\vec{J}$, as the sources of electromagnetic fields, are applied in the medium by using antennas and these can propagate large distances. If the medium is a free source or a partially free source, the value of $\vec{\rho}_v$ is zero and $\vec{J}$ is the induced conduction current.

2.2. Linearity, isotropy and homogeneity

For electrically homogeneous media it is implied that the electrical properties are independent of space. And, the assumption of isotropic properties infers that they are identical in all directions. The following constitutive relations are used to relate fluxes and field intensities in linear dielectrics, which for large electric fields remain dielectric and do not turn into conductors.

$$\vec{D} = \varepsilon_0 \varepsilon_r \vec{E} = \varepsilon \vec{E} \tag{5}$$

$$\vec{B} = \mu_0 (\vec{H} + \vec{M}) \tag{6}$$

where ε_0 and μ_0 are the electric permittivity and magnetic permeability of the vacuum or free space with values 8.854×10^{-12} F/m and $4\pi \times 10^{-7}$ H/m. The parameter ε_r is the electric relative permittivity of the medium. For simple homogeneous isotropic dielectric materials the magnetic permeability is constant. The quantity $\vec{M}$ in Eq. (6) is magnetization of the material, the average magnetic moment per unit volume. For non-magnetic materials (like diamagnetic and paramagnetic materials), which are the focus of this study, magnetization is zero or magnetic permeability remains like free space. However, for a porous medium that contains ferromagnetic particles such as Cobalt and Nickel, magnetization should be considered.

If it is assumed that an external source is applied at the boundary of the medium, the current inside charge free medium is the conduction current. The displacement current already has involved in Maxwell's equation. The current can be determined using

$$\vec{J} = \sigma \vec{E} \tag{7}$$

where σ is the medium conductivity in S/m.

2.3. Time harmonic fields

Maxwell's equations simplify considerably when the field that varies is periodic or sinusoidal with time. For antennas emitting high frequency waves this assumption seems reasonable. Therefore, the time dependency in the form of $e^{j\omega t}$ is defined where ω is the angular frequency in rad/s, t is time in seconds, and j is the imaginary unit. Using this time dependency function, each quantity can be introduced in the phasor (stationary) form. Using the assumptions of linear, isotropic, homogeneous, lossy dielectric, charge free, and in the phasor form the Maxwell's equations simplify to:

$$\nabla \cdot \vec{E}_s = 0 \tag{8}$$

$$\nabla \cdot \vec{H}_s = 0 \tag{9}$$

$$\nabla \times \vec{E}_s = -j\omega\mu \vec{H}_s \tag{10}$$

$$\nabla \times \vec{H}_s = (\sigma + j\omega\varepsilon)\vec{E}_s \tag{11}$$

and these equations can be further simplified by using the vector identity $\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$ to homogeneous vector Helmholtzs' equations or simply the vector wave equations;

$$\nabla^2 \vec{E}_s - \gamma_{em}^2 \vec{E}_s = 0 \tag{12}$$

$$\nabla^2 \vec{H}_s - \gamma_{em}^2 \vec{H}_s = 0 \tag{13}$$

where $\gamma_{em} = \alpha_{em} + j\beta_{em}$, in reciprocal of meter, is the propagation constant. Finally, α_{em} the attenuation factor (or absorption coeffi-

cient) and β_{em} the phase constant (or wavenumber) can be determined from the following relationships:

$$\alpha_{em} = \omega\sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right)} \tag{14}$$

$$\beta_{em} = \omega\sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}+1\right)} \tag{15}$$

2.4. Power dissipation

Poynting's theorem states that the net power flowing out of a given volume is equal to the time rate of the decrease in the energy stored minus the ohmic losses (Sadiku, 2014). It introduces Poynting vector ($\wp = \vec{E}\times\vec{H}$) measured in W/m^2 and in the direction of wave propagation. And, the time average Poynting vector over the period $\tau = 2\pi/\omega$ is

$$\wp_{ave} = \frac{1}{\tau}\int_0^{\tau}(\vec{E}\times\vec{H})dt = \frac{1}{2}Re(E_s\times H_s^*) \tag{16}$$

where H_s^* is the conjugate of H_s and, finally, the volumetric heat generation in W/m^3 can be obtained by

$$\dot{q}_{em} = -\nabla\cdot\wp_{ave} \tag{17}$$

Based on the mathematical models described here, the heat generation or the so called Beer-Lambert model for linear and radial coordinates are given, respectively, as (see Appendix A for more details).

$$\dot{q}_{em}(z) = \frac{2\alpha_{em}P_0}{A}e^{-2\alpha_{em}z} \tag{18}$$

$$\dot{q}_{em}(r) = \frac{2\alpha_{em}P_0}{2\pi h}\frac{e^{-2\alpha_{em}(r-r_w)}}{r} \tag{19}$$

The parameters z and r are the wave propagation direction, and P_0 is the amount of power which transmits into the porous media. In reality, there is a reflection when an electromagnetic wave front is at an interface between two different media. This reflection implies that some part of the wave returns into the medium from which it originated and only a fraction of the incident power penetrates into the desired medium. In this study, we assume that all the incident power at the inlet will pass into the porous medium. Later, we will show how the measured temperatures at the location of an antenna can be used to predict the actual heat absorbed by the media leading to an estimate of the heating efficiency of the process.

3. Heat conduction model

The heat generation as a result of electromagnetic wave propagation in a homogeneous, isotropic and infinite lossy porous medium was described in the previous section. In this section, we couple the heat generation model with transient heat conduction in a homogenous medium. Then, we obtain the analytical solutions for the transient heat conduction problems in Cartesian and radial coordinates subject to different boundary condition for a homogenous medium in the absence of phase change. Since we are interested in modeling of electromagnetic preheating of oil sands where phase change is not significant, the assumption of saturation independency of the absorption coefficient seems valid.

3.1. Linear case

We assume a one-dimensional medium with $0 < z < L$ (see Fig. 1a), which can be electrically semi-infinite provided that the length of the medium is much larger than the field penetration depth ($L \gg 1/\alpha_{em}$). This assumption for the application of interest, which is heating of geological media, seems reasonable. For an infinite case all of the applied energy is absorbed in the medium. However, for a finite domain the outer boundary can be assumed to be thermally insulated but allowing waves to pass.

The partial differential equation of heat conduction and high frequency electromagnetic heating in a linear system in the absence of fluid flow in dimensionless form is

$$\frac{d^2T_D}{dz_D^2} + \frac{1}{N}e^{-z_D} = \frac{\partial T_D}{\partial t_D}. \tag{20}$$

The term T_D is dimensionless temperature, z_D is dimensionless distance from the antenna, N is the Stark number, which is used to relate conduction-radiation parameters (Siegel et al., 2015), and t_D is dimensionless time. The medium temperature is initially constant ($T_D|_{t_D=0} = 0$). The boundary conditions of the system are determined by the symmetry at the inlet where the electromagnetic antenna is located and zero heat at the outer boundary (Carrizales et al., 2008).

$$\left.\frac{dT_D}{dz_D}\right|_{z_D=0} = \left.\frac{dT_D}{dz_D}\right|_{z_D=L_D} = 0 \tag{21}$$

The dimensionless parameters and reciprocal of N are defined as $T_D = (T-T_0)/T_0$, $z_D = 2\alpha_{em}z$, $t_D = 4k_T\alpha_{em}^2 t/M_t$, and $1/N = P_0/(2k_TT_0A\alpha_{em})$, where T is temperature in K, α_{em} is the absorption coefficient in reciprocal of meter, z is linear distance from the antenna in meter, k_T is the effective thermal conductivity in W/m K, M_t is the total volumetric heat capacity in J/m^3 K, P_0 is the incident power in W, and A is the cross-sectional area in m^2. Using the Fourier transform method, the solution of Eq. (20) can be obtained as

$$T_D = \frac{1}{NL_D}(1-e^{-L_D})t_D + \frac{2}{NL_D}\sum_{n=1}^{\infty}\frac{(1-(-1)^n e^{-L_D})}{(1+\lambda_n^2)\lambda_n^2}\left[1-e^{-\lambda_n^2 t_D}\right]\cos(\lambda_n z_D) \tag{22}$$

where $\lambda_n = n\pi/L_D$ is the eigenvalue of the corresponding boundary value problem. If the medium is unbounded ($0 < z < \infty$), the temperature at infinity remains at the initial condition, and the solution can be determined using the Laplace transform as

$$T_D = \frac{1}{2N}\left(\begin{array}{c} -2z_D erfc\left(\frac{z_D}{2\sqrt{t_D}}\right) - e^{t_D-z_D}erfc\left(\frac{z_D}{2\sqrt{t_D}}-\sqrt{t_D}\right) \\ +e^{t_D+z_D}erfc\left(\frac{z_D}{2\sqrt{t_D}}+\sqrt{t_D}\right) + \frac{4\sqrt{t_D}}{\sqrt{\pi}}e^{-\left(\frac{z_D}{2\sqrt{t_D}}\right)^2} + 2e^{t_D-z_D} - 2e^{-z_D} \end{array}\right) \tag{23}$$

3.2. Radial case

For the radial coordinate, we assume that the electromagnetic antenna is at the center of the cylinder and radiates outward (Fig. 1b). The domain extends from $r_w < r < r_e$ wherein r_w tends to zero (line-source approximation). The outer boundary can be either large enough to act as an infinite system or insulated. The governing equation in radial coordinates is given by:

$$\frac{1}{r_D}\frac{\partial}{\partial r_D}\left(r_D\frac{\partial T_D}{\partial r_D}\right) + \frac{1}{N}\frac{e^{-r_D}}{r_D} = \frac{\partial T_D}{\partial t_D} \tag{24}$$

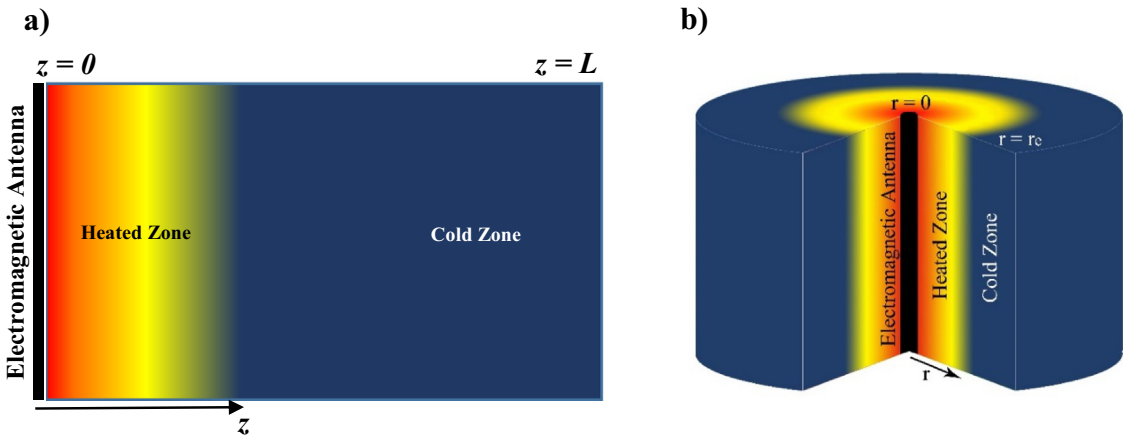


Fig. 1. Electromagnetic heating schematic for (a) linear, and (b) radial porous medium.

where the dimensionless radius and the Stark number in radial coordinates are $r_D = 2\alpha_{em} r$, and $1/N = P_0/(2\pi h k_T T_0)$, respectively. Herein h is the height of the cylinder and the rest of the parameters have been defined previously. The initial and boundary conditions are given by:

$$\left.\frac{\partial T_D}{\partial r_D}\right|_{r_{Dw}\to 0} = \left.\frac{\partial T_D}{\partial r_D}\right|_{r_D = r_{De}} = T_D|_{t_D=0} = 0 \tag{25}$$

The parameters r_{Dw} and r_{De} are dimensionless radii of the inner and outer boundaries, respectively. The analytical solution of Eq. (24) using the Hankel transform is

$$T_D(r_D, t_D) = \frac{2}{N r_{De}^2} \leq (1 - e^{-r_{De}}) t_D + \sum_{n=1}^{\infty} \frac{q_n}{\lambda_n^2}\left[1 - e^{-\lambda_n^2 t_D}\right] J_0(\lambda_n r_D) \tag{26}$$

where J_0 is the first-kind Bessel function, $\lambda_n = \alpha_n / r_{De}$, and α_n's are the roots of $J_1(\alpha_n) = 0$. The series parameter q_n can be determined from the following equation

$$q_n = \frac{2}{N r_{De}^2 J_0(\alpha_n)^2} \int_0^{r_{De}} e^{-r_D} J_0(\lambda_n r_D) dr_D \tag{27}$$

The integral shown by Eq. (27) can be determined numerically for a finite domain. However, when the outer radius is much larger than the wave penetration depth $(1/\alpha_{em})$, or in other words when $(r_{De} \gg 2)$, the integral can be approximated by Liu et al. (1999):

$$\int_0^{r_{De}} e^{-r_D} J_0(\lambda_n r_D) dr_D = \frac{1}{\sqrt{1+\lambda_n^2}} \tag{28}$$

And finally, the temperature profile for semi-infinite medium can be obtained as:

$$T_D(r_D, t_D) = \frac{2}{N r_{De}^2}(1 - e^{-r_{De}}) t_D + \frac{2}{N r_{De}^2} \sum_{n=1}^{\infty} \frac{J_0(\lambda_n r_D)}{J_0(\lambda_n)^2 \lambda_n^2 \sqrt{1+\lambda_n^2}} (1 - e^{-\lambda_n^2 t_D}) \tag{29}$$

Eq. (24) can also be solved when the outer boundary condition (Eq. (25)) is defined at infinity using the Laplace transform with respect to time and Hankel transform in r-direction. The solution in integral form is:

$$T_D(r_D, t_D) = \int_0^{\infty} \frac{1}{N\sqrt{1+u^2}\, u}\left[1 - e^{-u^2 t_D}\right] J_0(u r_D) du \tag{30}$$

It is worth noting that the above analytical solutions are developed using simplifying assumptions, which are explained earlier. In summary, these assumptions include time harmonic wave and lossy, linear, isotropic, homogeneous and semi-infinite medium. In addition, thermal conductivity of the medium is assumed to be constant. Phase change and dependency of the absorption coefficient on temperature are also neglected. For radial case line-source approximation is assumed.

4. Application to electromagnetic heating of oil sands

While the economics of conventional and shale oil production appear to tolerate current oil prices, the traditional thermal recovery methods of heavy oil and bitumen such as Steam Assisted Gravity Drainage (SAGD) seem to struggle. This is mostly due to high energy intensity of the current in situ thermal recovery processes where typically 3–6 volumes of steam are required to produce a unit volume of bitumen. This volume of steam translates into vast energy requirements. In addition, environmental regulations have motivated producers to seek recovery methods with less environmental impact. The high frequency electromagnetic waves as a source of power generation has motivated the Canadian government and industry to invest over 60 million dollars to pursue pilot scale demonstration of the technology ("Climate Change and Emissions Management (CCEMC) Corporation"). The high frequency electromagnetic heating is believed to be a clean technology that can potentially eliminate the need for steam with significantly less greenhouse gas emission. ESEIEH (Effective Solvent Extraction Incorporating Electromagnetic Heating) project with the objective of replacing steam with electromagnetic heating in combination with solvent dilution was launched in 2009 by the ESEIEH consortium. The first phase of the project was conducted at Suncor's North Steepbank mine to demonstrate the capability of the method for bitumen viscosity reduction (Trautman et al., 2014). The second phase of the project, which is a pilot at Suncor's Dover site, is currently underway and incorporates both radio frequency heating and low temperature solvent injection ("Suncor Energy, Technology Development").

In the following section we have tested the applicability of the developed analytical solutions to predict the temperature distribution around an antenna and the average temperature using the field data available in the literature (Ghannadi et al., 2014). Then, we used the developed analytical solution to obtain simple asymptotic solutions that allow prediction of the absorbed power by the geological media, which can be used to estimate efficiency of electromagnetic heating. In addition, the developed solutions are used to obtain a simple equation for estimation the electromagnetic absorption coefficient of geomaterials.

4.1. Temperature distribution

One of the concerns during heating of oil sands is coke formation during the process. The formation of coke affects the heating process and impairs the wellbores and antenna. Experimental data and modeling studies (Hassanzadeh et al., 2016) suggested a maximum operations temperature of 300 °C to minimize coke formation. The analytical solutions developed in the previous section are used to predict the temperature profiles around the antenna during EMH of oil sands and to investigate the possibility of coke formation.

The main electromagnetic property for the determination of the absorbed power by porous media from the electric and magnetic fields is absorption coefficient. Typical values of the absorption coefficient for the frequencies between 10 MHz and 1 GHz are computed for a geological medium based on the data reported in the literature (Chute and Vermeulen, 1990). The required data for Eq. (14) and the absorption coefficient values in SI units are shown in Table 1. A magnetic permeability of $4\pi \times 10^{-7}$ H/m is assumed in all cases. We assume that these properties are independent of temperature.

The absorption coefficient typically varies between 0.5 and 5 1/m for industrial high frequency heating. The thermal properties of the medium used in this study for both the linear and radial cases are also summarized in Table 1 (Ghannadi et al., 2014).

Fig. 2 shows the result of analytical solutions for temperature distribution during electromagnetic heating of both linear and radial systems. Maximum temperatures occur at the inlet (antenna location) and they increase as radiation time increases. In the radial case temperature decreases sharply moving away from the antenna. The logarithmic scales clearly demonstrate zero slope in the vicinity of the antenna.

The maximum dimensionless temperature at the location of antenna for the linear case at $t_D = 200$ ($t = 97.4$ days) is about 1.23 ($T = 370$ °C), which is higher than coke formation temperature. The antenna experiences coke formation temperature (~300 °C) after 64.6 days of heating. For the radial geometry, with the same amount of applied power (1000 W), inlet temperature reaches 214 °C ($T_D = 0.7$) after 97.4 days. The results show that the amount of power applied should be optimized based on the absorption coefficient and thermal parameters of geomaterials to avoid damage due to coke formation.

Thermal recovery of bitumen using SAGD involves a preheating period of 3–6 months where the formation around a pair of horizontal wells is preheated to a temperature such that the viscous bitumen becomes sufficiently mobile. Fig. 3a provides average temperature in the vicinity of a wellbore for different electromagnetic heating rates. These results provide an estimate of the average temperature around the wells during the startup period and a measure of bitumen mobility around the wellbore. Fig. 3b shows antenna temperature increases directly with applied power and frequency. If we suppose $t_D = 200$ (97.4 days) as the end of preheating period with the applied power of 1500 W, frequencies less than around 60 MHz are safe and impose no risk to the antenna.

4.2. Estimation of electromagnetic heating efficiency and the absorption coefficient

The temperature at the inlet (location of the antenna) can be obtained from Eq. (23) as

$$T_D|_{z_D=0} = \frac{1}{2N}\left(2e^{t_D}erfc(\sqrt{t_D}) + \frac{4\sqrt{t_D}}{\sqrt{\pi}} - 2\right) \tag{31}$$

At early time, the temperature at the location of the antenna follows a unit slope on log-log scale, that is $\lim_{t_D\to 0}\partial(\log T_D)/\partial(\log t_D) = 1$. It is worth noting that for constant heat flux, when all power is applied at the inner boundary and is transferred by pure conduction, the temperature at the inlet boundary can be obtained as $T_D(z_D = 0) = (2/N)\sqrt{t_D/\pi}$. Fig. 4 shows the inlet temperature versus time for two different Stark numbers of 12.5 and 15 (1000 and 810 W, respectively). In contrast with constant heat flux, where the inlet temperature at the early time varies with square root of time, the results show that temperature variations for electromagnetic heating are directly proportional to time of heating. However, at late time when heat conduction dominates, the temperature at the antenna approaches the constant heat flux case.

Normally, there is an impedance mismatch between transmission lines and antenna and also between the antenna and the oil sand formation. Therefore, all the energy converted to high frequency waves at the surface (i.e. theoretical energy) cannot be delivered to the new medium and some fraction is reflected back to generate the Voltage Standing Wave Ratio (VSWR). The VSWR varies between one and infinity and values close to one are preferable. In field applications, the length of antenna and the practical frequency have to be selected so that VSWR remains near one. Energy efficiency of high frequency heating is of great importance and its monitoring provides a useful means to control the heating process especially during the startup phase. The developed analytical solution can be used to estimate the power delivered to the oil sands during electromagnetic heating of oil sands.

For the late times, the electrical penetration depth is small as compared to the thermal penetration depth and the inlet temperature, which is described by $T_D(z_D = 0) = (2\sqrt{(t_D/\pi)} - 1)/N$, approaches the solution of constant flux for pure conduction heating at the inlet where $T_D(z_D = 0) = 2\sqrt{(t_D/\pi)}/N$. Since at the late time $\sqrt{(t_D/\pi)} \gg 0.5$, the temperature on a log-log scale shows a half slope behavior similar to the constant flux heating (see Fig. 4a).

In dimensional form, the late time temperature profile is given by:

Table 1
Electrical and thermal properties of a typical geological media.

Electrical properties					
Frequency	Units	10 MHz	200 MHz	500 MHz	1 GHz
Electrical conductivity	(S/m)	0.02	0.05	0.06	0.08
Electrical permittivity	(F/m)	1.328×10^{-10}	8.854×10^{-11}	8.854×10^{-11}	8.854×10^{-11}
Absorption coefficient	(1/m)	0.73	2.91	3.55	4.75
Thermal properties					
Thermal conductivity	1.45 (W/m K)		Cross sectional area		5 m^2
Thermal diffusivity	2.07×10^6 (J/m^3 K)		Length		20 m
Incident power	1000 W		Initial temperature		288 (K)
Outer radius (for radial)	10 m		Height (for radial)		2 m

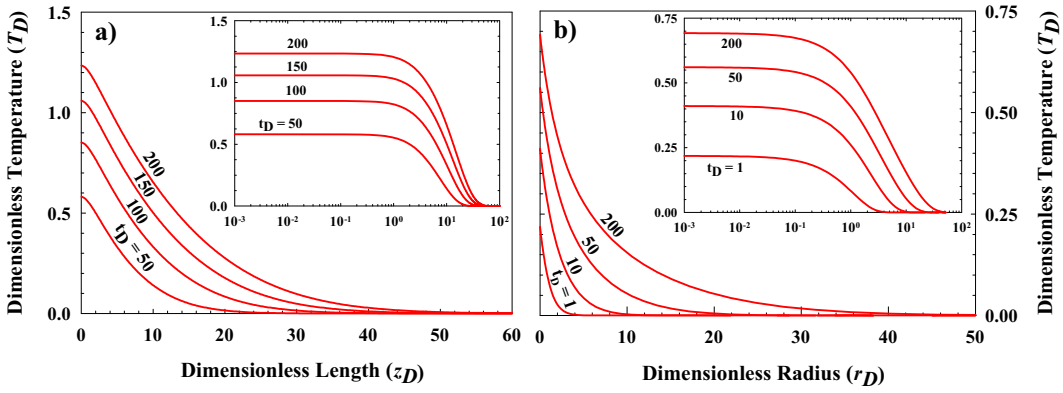


Fig. 2. Temperature profile for EMH by 200 MHz frequency (α_{em} = 2.91). (a) Linear system (N = 12.15) and (b) radial system (N = 5.25).

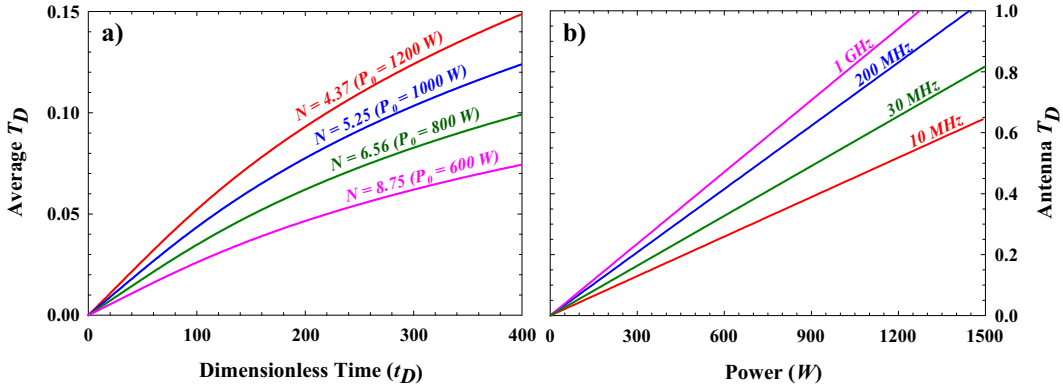


Fig. 3. (a) Average temperature for 5 m (the typical vertical distance between horizontal well-pairs of SAGD operation) around the antenna when it is radiating 200 MHz frequency waves and (b) antenna temperature variation with applied power and frequency at t_D = 200 (97.4 days).

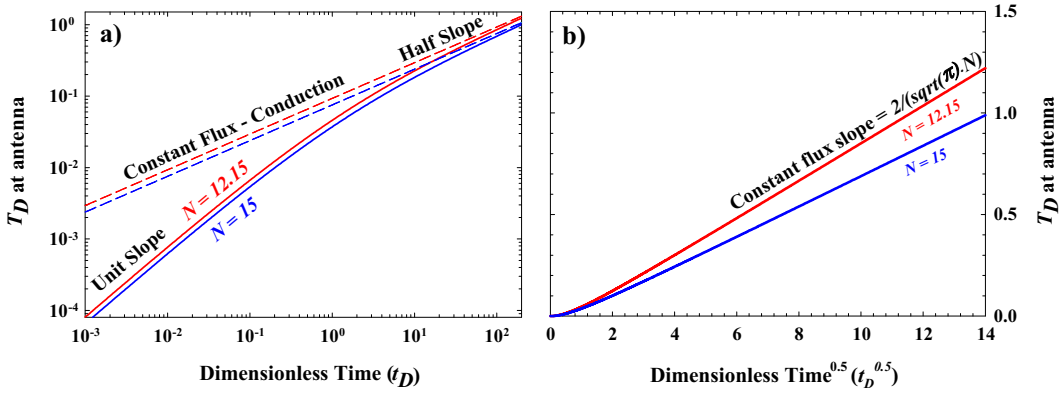


Fig. 4. Temperature at the antenna versus time when the applied frequency is 200 MHz (α_{em} = 2.91) and N = 12.15 and 15. (a) Log-log plot to show the slopes at early- and late-times (b) the linear slope at the late times.

$$T = \frac{2P_0}{A\sqrt{\pi k_T M_t}}\sqrt{t} + T_0 - \frac{P_0}{2k_T A\alpha_{em}}, \tag{32}$$

and consequently, the data of electromagnetic heating process when plotted versus square root of time gives slope $m = 2P_0/(A\sqrt{\pi k_T M_t})$ and the intercept $I = T_0 - P_0/(2k_T A\alpha_{em})$. The slope can be used to estimate the absorbed power as given by $P_0 = mA\sqrt{\pi k_T M_t}/2$ and then the intercept and slope can be used to estimate the absorption coefficient as given by $\alpha_{em} = m\sqrt{\pi M_t}/(4\sqrt{k_T}(T_0 - I))$. These parameters are of great importance in design of experiments and field applications of electromagnetic heating of oil sands.

Fig. 5 shows the inlet temperature versus time for the radial case for two different Stark numbers of 5.25 and 10.5 (1000 and 500 W, respectively). At early time, the temperature at the location of the antenna shows unit slope on log-log scale, while constant heat flux line-source problem gives zero slope (Sabet, 1991). Similar to the linear case at the late time, we expect the behavior of EM radiation to be close to pure conduction constant flux case with line-source approximation, which is given by $T_D(r_D = 0, t_D) = (\ln(t_D) + 0.809)/2N$. Since Eq. (30) is complex, it

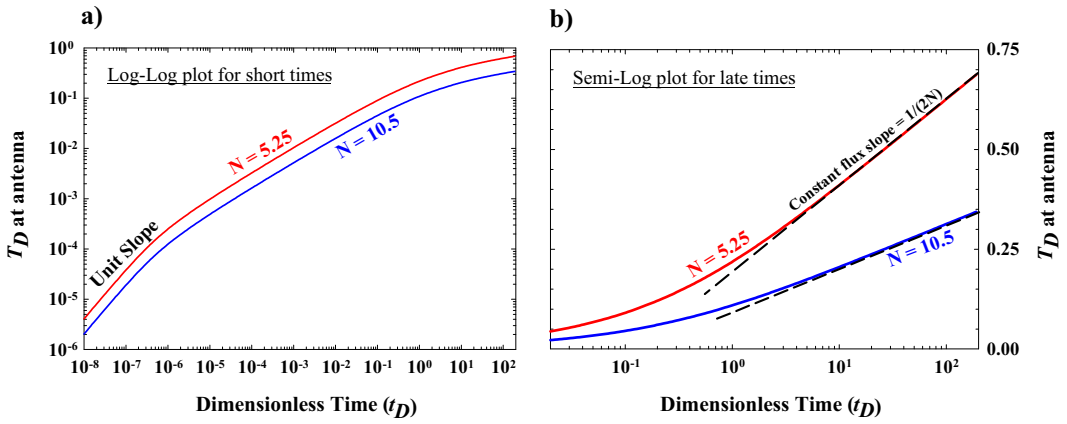


Fig. 5. Temperature profile for radial EMH at antenna location for 200 MHz frequency and N = 5.25 and 10.5. (a) Log-log plot to show the early-time period and (b) semi-log plot for determination of N at late times.

is graphically determined that the dimensionless temperature at the location of antenna at the late time follows $T_D(r_D = 0, t_D) = (\ln(t_D) + 1.954)/2N$. The graphical representation of this solution is shown in Fig. 6.

Late time temperature data when plotted versus time on a semi-log scale in dimensional form gives a line with slope of $m = P_0/(4\pi hk_T)$. The intercept at t = 1 h, chosen to simplify calculations, is given by:

$$I = T(t = 1\text{ h}) = T_0 + \frac{P_0}{(4\pi hk_T)}(\ln(14400k_T\alpha_{em}^2/M_t) + 1.954). \quad (33)$$

The slope can be used to estimate the absorbed power as given by $P_0 = m(4\pi hk_T T_0)$ and the intercept can be used to determine the absorption coefficient after some simplification. Estimation of these parameters is of great value in field application of electromagnetic heating.

4.3. Thermal penetration depth of EM heating

Several definitions have been suggested in the literature to estimate the penetration depth of a diffusive front (Jimenez et al., 2013). One approach is to find the depth at which $\partial(\partial T_D/\partial z_D)/\partial t_D$ for the linear case and $\partial(\partial T_D/\partial \ln(r_D))/\partial \ln(t_D)$ for the radial case are maximized (Thompson and Reynolds, 1997). Figs. 7a and 8a show these terms versus distance for linear and radial cases, respectively, which show increase of penetration depth with time, as expected. Figs. 7b and 8b show the comparison between the penetration depth with different frequencies and also the pure conduction without radiation case with a constant heat flux. Note that in the pure conduction case the penetration depth is independent of the input power and it behaves linearly when plotted versus square root of time. At higher frequencies the absorption coefficients are higher leading to absorption of the incident power in the vicinity of the antenna while at lower frequencies the incident power penetrates deep into the medium resulting in larger penetration depth. Away from the antenna the dominant mechanism of heat transport is thermal conduction and the penetration depth versus time approaches the pure conduction case. The results show that at higher frequencies the electromagnetic heating approaches conductive heating with a constant heat flux in a shorter period of time.

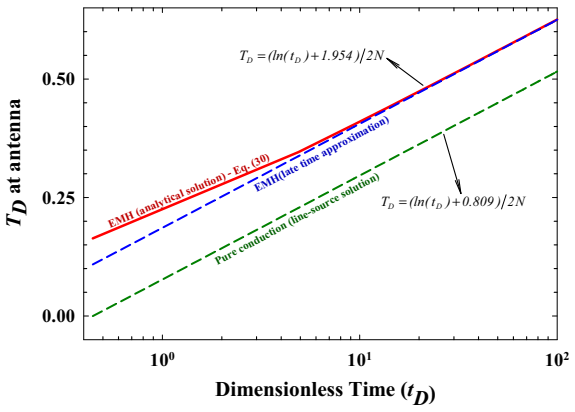


Fig. 6. Late time behavior of temperature profile for radial EMH at the location of antenna for 200 MHz frequency and N = 5.25.

5. Summary and conclusion

Current in-situ thermal recovery methods such as Steam Assisted Gravity Drainage (SAGD) struggle to be economic due to high energy intensity and recent low oil prices. Electromagnetic heating of oil sands has been suggested for in situ delivery of thermal energy to viscous bitumen thereby improving the heating efficiency and reducing environmental impact. In this study, we used the Beer-Lambert law formulation to couple heat generation as a result of electromagnetic wave propagation in porous media with thermal conduction. New analytical solutions for electromagnetic heating of lossy geological media were developed in Cartesian and radial coordinates with application to bitumen extraction from oil sands. The new analytical solutions can be used as a forward model in design of experimental studies of electromagnetic heating of geomaterials and to predict the electromagnetic absorption coefficient. Asymptotic solutions were developed that can be used in field application to estimate power loss as a result of impedance mismatch between antenna and geologic media and the absorption coefficient of the geological materials. The solutions presented here also provide a simple tool to predict temperature distribution during electromagnetic heating of solid materials. These results find applications in design of experiments at pilot and field scale for implementation of electromagnetic heating of oil sands.

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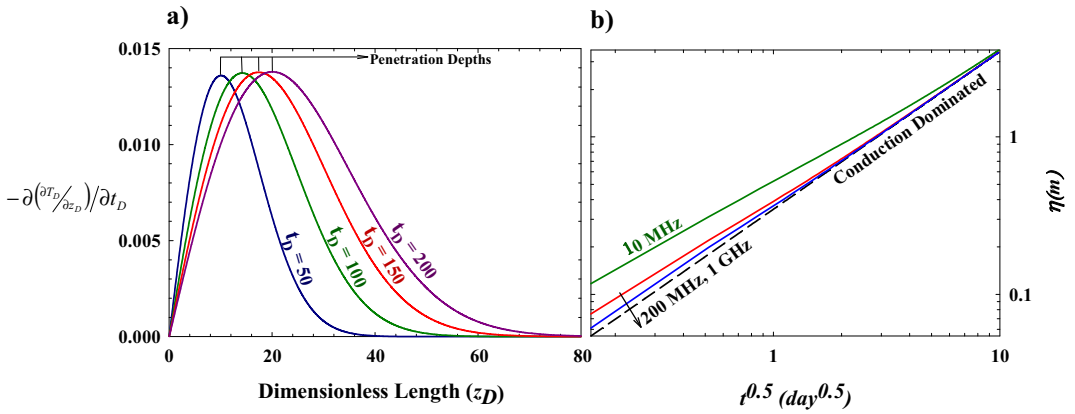


Fig. 7. (a) Calculation of penetration depth in linear case for different times at the frequency of 200 MHz and (b) Linear penetration depth changes with time for different frequencies.

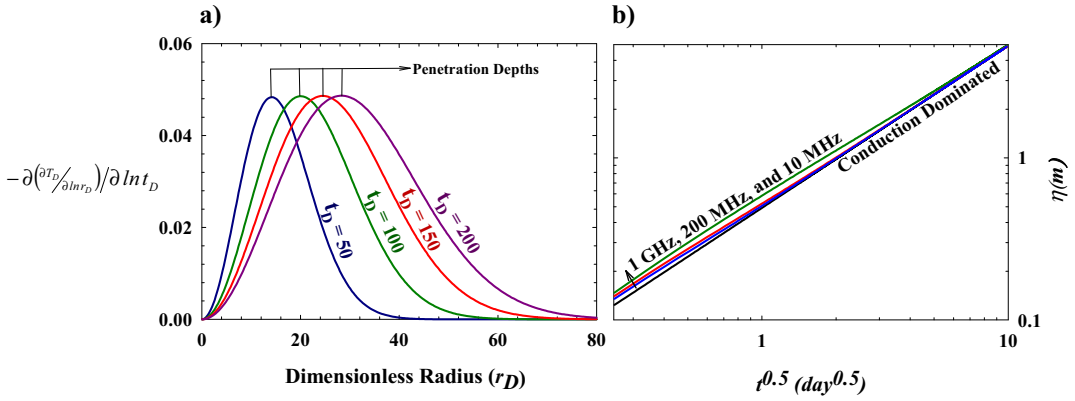


Fig. 8. (a) Calculation of penetration depth in radial case for different times at the frequency of 200 MHz and (b) radial penetration depth changes with time for different frequencies.

Appendix A. Wave propagation in lossy media

A.1. Cartesian systems

If we assume that a current is varying sinusoidally in x-direction, it generates the magnetic field normal to the x-z plane (y-direction), which through the Faraday's law leads to the electric field in the x-direction and it again results in another magnetic field away from the antenna. Solving the wave equation (Eq. (12)) gives $E_{xs}(z) = E_0 e^{-\gamma_{em} z}$, and then by inserting the time factor, the electric field intensity is obtained as:

$$\vec{E} = E_0 e^{-\alpha_{em} z} \cos(\omega t - \beta_{em} z)\vec{a}_x \tag{A1}$$

Using Eq. (A1) in conjunction with Eq. (10), applying time factor, and introducing the intrinsic impedance of the media ($\eta = E_0/H_0$), in ohms, the magnetic field intensity can be computed by

$$\vec{H} = \frac{E_0}{|\eta|} e^{-\alpha_{em} z} \cos(\omega t - \beta_{em} z - \theta_\eta)\vec{a}_y. \tag{A2}$$

The wave propagation schematic is shown in Fig. A1.The exponential terms show that the fields are decreasing as they propagate along the z-direction with a coefficient α_{em}, hence it is known as the absorption coefficient. In Eq. (A2) θ_η is impedance phase. The complex intrinsic impedance of media equals (Sadiku, 2014):

$$\eta = |\eta| e^{j\theta_\eta} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} \tag{A3}$$

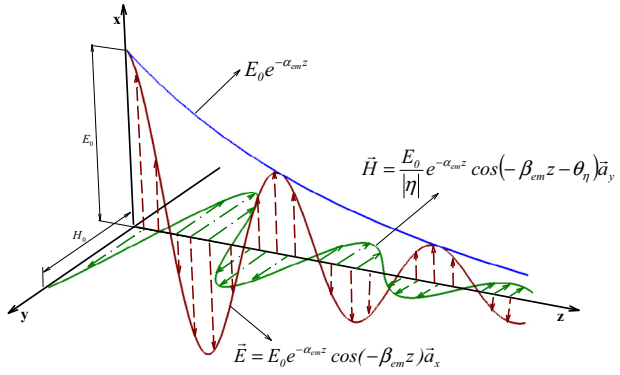


Fig. A1. The electric and magnetic wave propagation in lossy media with 30° impedance phase.

for which their magnitude and phase are $|\eta| = \sqrt{\mu}/(\sqrt{\varepsilon}(1 + (\sigma/\omega\varepsilon)^2)^{1/4}$ and $\theta_\eta = 0.5 \arctan(\sigma/\omega\varepsilon)$, respectively. Eqs. (A1) and (A2) express that $\vec{E}$ and $\vec{H}$ are out of phase by angle θ_η due to the intrinsic impedance of the medium. Impedance phase shows how a medium is lossy; for highly lossy media like conductors it is near 45° and for perfect dielectrics is zero.

For determination of the thermal energy generation we use average Poynting vector (Eq. (16)) as

$$\wp_{ave}(z) = \frac{1}{2} Re(E_s \times H_s^*) = \frac{E_0^2}{2|\eta|} e^{-2\alpha_{em} z} \cos\theta_\eta \vec{a}_z, \tag{A4}$$

where $\vec{a}_z$ is the unit vector of wave propagation direction. The impedance magnitude and phase is dependent on constitutive parameters and applied frequency which is assumed to be constant. We substitute the effective energy flux (P_0/A) at the incident face ($z = 0$) instead of constant terms in Eq. (A4). Thus, the average Poynting vector is obtained as:

$$\wp_{ave}(z) = \frac{P_0}{A} e^{-2\alpha_{em} z} \vec{a}_z, \tag{A5}$$

where P_0 is the effective applied power (or absorbed power). Consequently, the volumetric heat dissipation of electromagnetic heating in (W/m^3) for a linear system is obtained as:

$$\dot{q}_{em}(z) = \frac{2\alpha_{em} P_0}{A} e^{-2\alpha_{em} z} \tag{A6}$$

A.2. Radial systems

For the radial case, the magnetic field is circulating tangent to the azimuthal angle direction and the electric field is in the z-direction. Therefore, the wave propagates radially away from the antenna. Using Helmholtzs' equations (Eqs. (12) and (13)) to solve the electric and magnetic field intensities lead to Hankel's Bessel function of the first kind as fields are converging for large radii.

$$\vec{E}_{zs}(r) = E_0 H_0^1(j\gamma_{em} r)\vec{a}_z \tag{A7}$$

$$\vec{H}_{\phi s}(r) = \frac{E_0}{\eta} H_0^1(j\gamma_{em} r)\vec{a}_\phi \tag{A8}$$

From the asymptotic form of the Hankel functions as $j\gamma_{em} r \to \infty$, approximation can be made as (Jensen et al., 2011):

$$H_0^1(j\gamma_{em} r) \approx \sqrt{\frac{2}{\pi j\gamma_{em} r}} e^{-\gamma_{em} r - j\pi/4}. \tag{A9}$$

Then, the average Poynting vector becomes

$$\wp_{ave}(r) = \frac{1}{2} Re(E_{zs} \times H_{\phi s}^*) = \frac{E_0^2}{\pi|\eta||\gamma_{em}|} \frac{e^{-2\alpha_{em} r}}{r} \cos\theta_\eta \vec{a}_r, \tag{A10}$$

where $\vec{a}_r$ is the unit vector of wave propagation direction. Like the linear case, the effective flux entering the media $P_0/(2\pi r_w h)$ changes Eq. (A10) to

$$\wp_{ave}(r) = \frac{P_0}{2\pi h} \frac{e^{-2\alpha_{em}(r-r_w)}}{r} \vec{a}_r, \tag{A11}$$

and finally, the volumetric dissipated power yield by

$$\dot{q}_{em}(r) = \frac{2\alpha_{em} P_0}{2\pi h} \frac{e^{-2\alpha_{em}(r-r_w)}}{r} \vec{a}_r. \tag{A12}$$

References

Abernethy, E.R., 1976. Production increase of heavy oils by electromagnetic heating. J. Can. Pet. Technol. 15. http://dx.doi.org/10.2118/76-03-12.

Ayappa, K.G., Davis, H.T., Crapiste, G., Davis, E.A., Gordon, J., 1991a. Microwave heating: an evaluation of power formulations. Chem. Eng. Sci. 46, 1005–1016. http://dx.doi.org/10.1016/0009-2509(91)85093-D.

Ayappa, K.G., Davis, H.T., Davis, E.A., Gordon, J., 1991b. Analysis of microwave heating of materials with temperature-dependent properties. AIChE J. 37, 313–322. http://dx.doi.org/10.1002/aic.690370302.

Basak, T., Ayappa, K.G., 1997. Analysis of microwave thawing of slabs with effective heat capacity method. AIChE J. 43, 1662–1674. http://dx.doi.org/10.1002/aic.690430703.

Bhattacharya, M., Venkata, S.K., Basak, T., 2013. Role of microwave heating strategies in enhancing the progress of a first-order endothermic reaction. AIChE J. 59, 656–670. http://dx.doi.org/10.1002/aic.13848.

Carrizales, M.A., Lake, L.W., Johns, R.T., 2008. Production improvement of heavy-oil recovery by using electromagnetic heating. In: SPE Annual Technical Conference and Exhibition. Society of Petroleum Engineers. http://dx.doi.org/10.2118/115723-MS.

Chute, F.S., Vermeulen, F.E., 1990. Physical modeling of systems involving electromagnetic heating of materials. IEEE Trans. Educ. 33, 190–197. http://dx.doi.org/10.1109/13.54860.

Climate Change and Emissions Management (CCEMC) Corporation [WWW Document], n.d. URL: <http://ccemc.ca/project/enhanced-solvent-extraction-incorporating-electromagnetic-heating/> (accessed 9.26.16).

Erchiqui, F., 2013. 3D numerical simulation of thawing frozen wood using microwave energy: frequency effect on the applicability of the Beer-Lambert law. Dry. Technol. 31, 1219–1233. http://dx.doi.org/10.1080/07373937.2013.784327.

Fanchi, J.R., 1990. Feasibility of reservoir heating by electromagnetic irradiation. In: SPE Annual Technical Conference and Exhibition. Society of Petroleum Engineers. http://dx.doi.org/10.2118/20483-MS.

Ghannadi, S., Irani, M., Chalaturnyk, R.J., 2014. Induction and radio frequency heating strategies for steam-assisted gravity drainage start-up phase. In: SPE Heavy Oil Conference-Canada. Society of Petroleum Engineers. http://dx.doi.org/10.2118/170037-MS.

Hassanzadeh, H., Harding, T., 2016. Analysis of conductive heat transfer during in-situ electrical heating of oil sands. Fuel 178, 290–299. http://dx.doi.org/10.1016/j.fuel.2016.03.070.

Hassanzadeh, H., Harding, T.G., Moore, R.G., Mehta, S.A., Ursenbach, M.G., 2016. Gas generation during electrical heating of oil sands. Energy Fuels 30, 7001–7013. http://dx.doi.org/10.1021/acs.energyfuels.6b01227.

Hippel, A.R..Von., 1954. Dielectrics and Waves. John Wiley and Sons, New York.

Hossan, M.R., Dutta, P., 2012. Effects of temperature dependent properties in electromagnetic heating. Int. J. Heat Mass Transf. 55, 3412–3422. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2012.02.072.

Jensen, F.B., Kuperman, W.A., Porter, M.B., Schmidt, H., 2011. Computational Ocean Acoustics. Springer, New York, New York, NY. http://dx.doi.org/10.1007/978-1-4419-8678-8.

Jimenez, M., Dietrich, N., Cockx, A., Hébrard, G., 2013. Experimental study of O_2 diffusion coefficient measurement at a planar gas-liquid interface by planar laser-induced fluorescence with inhibition. AIChE J. 59, 325–333. http://dx.doi.org/10.1002/aic.13805.

Keangin, P., Rattanadecho, P., Wessapan, T., 2011. An analysis of heat transfer in liver tissue during microwave ablation using single and double slot antenna. Int. Commun. Heat Mass Transf. 38, 757–766. http://dx.doi.org/10.1016/j.icheatmasstransfer.2011.03.027.

Lei, K.F., Ahsan, S., Budraa, N., Li, W.J., Mai, J.D., 2004. Microwave bonding of polymer-based substrates for potential encapsulated micro/nanofluidic device fabrication. Sensors Actuators A Phys. 114, 340–346. http://dx.doi.org/10.1016/j.sna.2003.12.018.

Liu, J., Spiegel, M.R., Spiegel, M.R., 1999. Mathematical Handbook of Formulas and Tables. McGraw-Hill.

Rahbar, M., Chhina, S., Sameoto, D., Parameswaran, M., 2010. Microwave-induced, thermally assisted solvent bonding for low-cost PMMA microfluidic devices. J. Micromech. Microeng. 20, 15026. http://dx.doi.org/10.1088/0960-1317/20/1/015026.

Rattanadecho, P., Keangin, P., 2013. Numerical study of heat transfer and blood flow in two-layered porous liver tissue during microwave ablation process using single and double slot antenna. Int. J. Heat Mass Transf. 58, 457–470. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2012.10.043.

Sabet, M.A., 1991. Well Test Analysis. Gulf Pub. Co..

Sadiku, M.N.O., 2014. Elements of Electromagnetics. Oxford University Press, New York, NY.

Siegel, R., Howell, J.R., Mengüç, M.P., 2015. Thermal Radiation Heat Transfer. CRC Press, New York, NY.

Suncor Energy, Technology Development [WWW Document], n.d. URL: <http://sustainability.suncor.com/2015/en/vision/technology-development.aspx> (accessed 9.26.16).

Thompson, L.G., Reynolds, A.C., 1997. Well testing for radially heterogeneous reservoirs under single and multiphase flow conditions. SPE Form. Eval. 12, 57–64. http://dx.doi.org/10.2118/30577-PA.

Trautman, M., Macfarlane, B., Martindale, T., 2014. Experimental and numerical simulation results from a radio frequency heating test in native oil sands at the North Steepbank Mine. In: World Heavy Oil Conference, New Orleans, Louisiana, USA.

Yang, H.W., Gunasekaran, S., 2004. Comparison of temperature distribution in model food cylinders based on Maxwell's equations and Lambert's law during pulsed microwave heating. J. Food Eng. 64, 445–453. http://dx.doi.org/10.1016/j.jfoodeng.2003.08.016.